

1. Evaluate the following limits. Write ∞ , $-\infty$ or DNE , as appropriate.

(a) (3 points) $\lim_{x \rightarrow 5^+} \frac{|5 - x|}{x^2 - 9x + 20}$

(b) (3 points) $\lim_{x \rightarrow -\infty} \frac{\arctan(2x)}{e^{6x}}$

(c) (3 points) $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 + x}}{x}$

(d) (3 points) $\lim_{x \rightarrow \infty} \frac{(2x + 3)^3}{x^3 + 8}$

(e) (2 points) $\lim_{x \rightarrow 0} \left(\frac{\sin x}{3x} + \arccos(2x) \right)$

2. (4 points) Find all pairs of values (a, b) for which the following function is continuous for all x .

$$g(x) = \begin{cases} 1 + 4x - x^2 & \text{if } x < a \\ b & \text{if } x = a \\ (x - 3)^2 & \text{if } x > a \end{cases}$$

3. (4 points) Use the *limit definition of the derivative* to find $f'(x)$ for the function $f(x) = \frac{4}{3x + 1}$.

4. Find $\frac{dy}{dx}$ in each case. Do not simplify your answer.

(a) (4 points) $y = \left(\frac{3x + 2}{x^2 - 3} \right)^5 + 4^{3x}$

(b) (4 points) $y = e^{x^3 \tan x} + \arccos(2x)$

(c) (4 points) $y = \sin^5(\sqrt{x^2 + 1})$

(d) (4 points) $y = (\cos x)^{\ln x}$

5. (4 points) Find the values of x for which the tangent line to the curve $y = (x - 3)(3x + 5)^3$ is horizontal.

6. (5 points) Consider the function defined by the equation $x^4 + y^4 = 2xy^3 + 1$.

(a) Find $\frac{dy}{dx}$ by implicit differentiation.

(b) Find the equation of the tangent line at the point $(1, 2)$.

7. (6 points) A drone is flying straight upward while a camera on the ground tracks it. The camera is located 8 m from the point on the ground directly below the drone. If the drone is rising at a rate of 2 m/s, determine :

(a) the rate at which the distance between the camera and the drone is changing when the drone is 6 m high.

(b) the rate at which the camera's angle of elevation is changing when the drone is 8 m high.

8. (10 points) Consider the following function, along with its first and second derivatives.

$$f(x) = \frac{9(x - 1)^2}{(x + 5)^2}, \quad f'(x) = \frac{108(x - 1)}{(x + 5)^3}, \quad f''(x) = \frac{216(4 - x)}{(x + 5)^4}$$

- (a) Find the domain and intercepts of f .
- (b) Find the vertical and horizontal asymptotes of f (if any).
- (c) Find the intervals of increase/decrease of f .
- (d) Find the local (relative) extrema of f .
- (e) Find the intervals of concavity of f .
- (f) Find all points of inflection of f .
- (g) On the next page, sketch the graph of f .
9. (6 points) Find the radius and height of the cylinder whose total surface area is 600π units square for which the volume is maximized.
10. Evaluate the following integrals.
- (a) (4 points) $\int \frac{(x+3)^2}{2x\sqrt{x}} dx$
- (b) (4 points) $\int \tan x \left(\tan x + \frac{\cos x}{\sin^3 x} \right) dx$
- (c) (4 points) $\int_1^{\sqrt{3}} \left(\frac{1}{x} - \frac{6}{x^2+1} \right) dx$
11. (4 points) A particle moves in a straight line and has acceleration given by $a(t) = e^t - t + \cos t$. Its initial velocity is $v(0) = 5m/s$ and its initial position is $s(0) = 3m$. Find its position function $s(t)$.
12. (5 points) Evaluate $\int_0^2 (2 + 3x - x^2) dx$ as a limit of a Riemann sums. The following identities may be useful.
- $$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$
13. (3 points) Given that $\int_1^5 f(x) dx = 9$ and $\int_3^5 f(x) dx = 4$, find $\int_1^3 (3f(x) + 4) dx$.
14. (4 points) Find the area enclosed between the curves $y = (x-2)^2$ and $y = 4-x$.
15. (3 points) Let $g(x) = \int_1^x \frac{1}{1+t^4} dt$. Find an equation of the tangent line to the curve $y = g(x)$ at $x = 1$.

Answers

1.

- (a) 1
- (b) $-\infty$
- (c) $-1/2$
- (d) 8
- (e) $\pi/2 + 1/3$

2. $(a, b) = (1, 4)$ or $(a, b) = (4, 1)$.

$$3. f'(x) = \lim_{h \rightarrow 0} \frac{\frac{4}{3(x+h)+1} - \frac{4}{3x+1}}{h} = \dots = \frac{-12}{(3x+1)^2}.$$

4.

- (a) $\frac{dy}{dx} = 5 \left(\frac{3x+2}{x^2-3} \right)^4 \frac{3(x^2-3) - (3x+2)2x}{(x^2-3)^2} + 4^{3x} \cdot 3 \ln 4$
- (b) $\frac{dy}{dx} = e^{x^3 \tan x} (3x^2 \tan x + x^3 \sec^2 x) - \frac{2}{\sqrt{1-4x^2}}$
- (c) $\frac{dy}{dx} = 5 \sin^4(\sqrt{x^2+1}) \cos(\sqrt{x^2+1}) \frac{1}{2\sqrt{x^2+1}} \cdot 2x$
- (d) $\frac{dy}{dx} = (\cos x)^{\ln x} \left(\frac{\ln(\cos x)}{x} - (\ln x) \tan x \right)$

5. $x = -5/3, x = 11/6$

6.

- (a) $\frac{dy}{dx} = \frac{y^3 - 2x^3}{y^2(2y - 3x)}$
- (b) $y - 2 = \frac{3}{2}(x - 1)$

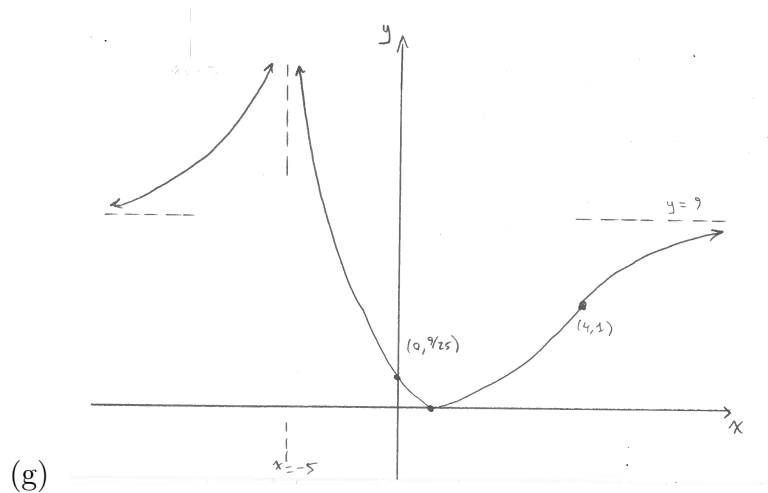
7.

- (a) The distance between the camera and the drone is increasing at the rate of $6/5$ m/s.
- (b) The camera's angle of elevation is increasing at the rate of $1/8$ rad/s.

8.

- (a) Domain : $\mathbb{R} \setminus \{-5\}$, x -intercept : $(1, 0)$, y -intercept : $(0, 9/25)$.
- (b) Vertical asymptote : $x = -5$, Horizontal asymptote : $y = 9$.
- (c) f increases on $(-\infty, -5)$ and on $(1, \infty)$. f decreases on $(-5, 1)$.

- (d) The point $(1, 0)$ is a relative minimum of f .
- (e) f is concave upwards on $(-\infty, -5)$ and on $(-5, 4)$. f is concave downwards on $(4, \infty)$.
- (f) The point $(4, 1)$ is a point of inflection of f .



9. The radius must be of 10 units and the height must be of 20 units.

10.

- (a) $\frac{1}{3}x\sqrt{x} + 6\sqrt{x} - \frac{9}{\sqrt{x}} + C$
- (b) $\tan x - x - \cot x + C$
- (c) $\frac{1}{2} \ln 3 - \frac{\pi}{2}$

11. $s(t) = e^t - \frac{1}{6}t^3 - \cos t + 4t + 3$

12. $22/3$

13. $9/2$

14. 23

15. $y = \frac{1}{2}(x - 1)$